# Exercises

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## Exercise 1.

A topological space is said to be *Hausdorff* if for any two distinct points in the space, there exists two disjoint open sets containing each point. Every metric space is automatically a Hausdorff space (see if you can prove this for yourself), but plenty non-Hausdorff spaces exist.

### (a) (Difficulty 2/10)

Show that if a finite group $G$ acts on a Hausdorff space $Y$ such that no nontrivial elements has a fixed point, the action is even. Recall that this means that for all $y \in Y$, there exists an open set $U$ containing $y$ such that the open sets $\{gU\}_{g \in G}$ are *pairwise disjoint*, i.e. $g_1 U \cap g_2 U = \varnothing$ for all $g_1, g_2 \in G$. By the result in lecture, this implies that if $Y$ is connected then $Y \to G \setminus Y$ is a Galois cover.

If you are not comfortable with working with topological spaces, you may assume that $Y$ is a metric space instead.

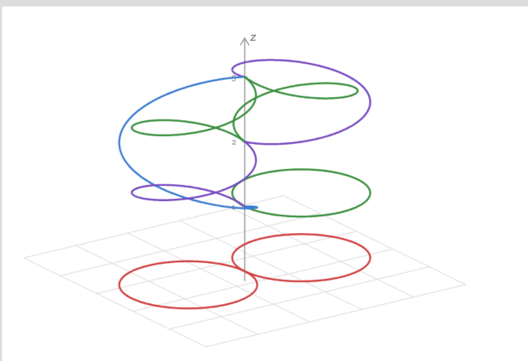
### (b) (Difficulty 4/10)

Show that the finiteness of $G$ is a necessary requirement, by considering the example where $Y = S^1 = \{(\cos(t), \sin(t)) \in \mathbb{R}^2 \mid t \in [0, 2\pi)\}$ is the unit circle and $G = \mathbb{Z}$; where $n \in \mathbb{Z}$ acts on $Y$ via rotation by $n$ radians counterclockwise; i.e. $n \cdot (\sin(t), \cos(t)) = (\sin(n+t), \cos(n+t))$. You need not verify that this is a well-defined group action, and you can take for granted the group also acts by *isometries*; i.e. $d(p,q) = d(n \cdot p, n \cdot q)$ for all $p, q \in S^1$ and $n \in \mathbb{Z}$.

## Exercise 2.

### (Difficulty 4/10)

Show that the following cover of the *figure 8* (denoted $S_1 \vee S_1$) is *not* Galois:



**Exercise 3.**

**(Difficulty 6/10)**

Show that a connected cover $Y \to X$ is Galois with automorphism group $G$ if and only if the fibre product $Y \times_X Y$ is isomorphic to the trivial cover $Y \times G \to Y$; where $G$ is assumed to carry the discrete topology.

**Exercise 4.**

**(Difficulty 10/10)**

Let $S^1 = \{(\cos(t), \sin(t)) \in \mathbb{R}^2 \mid t \in [0, 2\pi)\}$ denote the unit circle.

Show that every cover $p : S^1 \to S^1$ is Galois, and find its Galois group.

(If you are not comfortable with working with topological spaces, you may assume that every cover is already a metric space.)

(**Hint.** There are many ways to approach this problem. I outline one of them below, but you are welcome to try and figure out your own method.

(1) Suppose that $p : S^1 \to S^1$ is some degree $n$ cover. Construct a parameterization $f : [0, 2n\pi) \to S^1$ such that $(p \circ f)(t) = (\cos(t), \sin(t))$.

(2) Show that $f(t) \mapsto f(t + 2\pi)$ induces a covering automorphism of $S^1$ over $S^1$.

(3) Show that there are exactly $n$ distinct automorphisms, and how this implies that the cover is Galois. What is the group structure? )

**If you have only listened to my first talk, do *not* attempt the questions after this page** (you're welcome to take a look - they will just be nigh impossible to do without the theorems I will introduce in the second talk)**.**

**The second half of this document is meant to be done only after the second talk.**

**Warning:** I have tried my best to keep the questions at a manageable difficulty level for those with a minimal mathematical background. Despite my best efforts, the rest of this document represents a *significant* difficulty jump in the exercises. You are regardless welcome to attempt them and ask me questions if you need help.

**Exercise 5.**

**(a) (Difficulty 16/10)**

Let $X$ be a connected and locally path-connected, but not necessarily locally simply connected topological space. Show that every finite cover of $X$ is *dominated* by some Galois cover; i.e. for every finite cover $\pi : Y \to X$ there exists a cover $Z \to Y \to X$ such that $Z \to X$ is Galois.

(**Hint.** Prove the question statement via the following steps:

(1) First argue why you can WLOG assume that $Y$ is connected.

(2) Consider the $n$-fold fibered product of $Y$ over $X$, denoted $Y_X^n$. Notice that this space admits a natural action of $S_n$.

(3) Let $\hat{Y}$ denote the connected component of $Y_X^n$ that contains the point $(y_1, ..., y_n)$, where $\{y_i\} = \pi^{-1}(x)$ is the fiber above some chosen basepoint $x$. This space still admits the same action of $S_n$.

(4) Notice that $\pi_1(X, x)$ embeds as a transitive subgroup of $S_n$ via the monodromy action, and $S_n$ has a well-defined morphism into $\mathrm{Aut}\big(\hat{Y}|X\big)$. Argue why this implies that $\hat{Y}$ is Galois. )

**(b) (Difficulty 16/10)**

Show that the Galois cover you have constructed is *terminal* among all the Galois covers of $Y$ over $X$: Namely, if $W \to Y$ is any Galois cover, then there exists a unique morphism $W \to Z$ such that the following diagram commutes:

$$\begin{tikzcd}
W \arrow[d, dashed] \arrow[dr] & & \\
Z \arrow[r] & Y \arrow[r, "\pi"] & X
\end{tikzcd}$$

(this is an instance of a *universal property*.)

Hence, we see that every finite cover of a space admits a *Galois closure*, which corresponds to the similar notion that we have for finite separable field extensions.

**Exercise 6.**

A *topological group* is a group $G$ that is also a topological space, such that the group structure is compatible with the topology in the following sense: 1) group multiplication is a continuous function $G \times G \to G$ (where $G \times G$ is viewed with the product topology), and 2) inversion is a continuous function $G \to G$.

Examples of topological groups include $\mathbb{R}^n$ under addition, or the circle group $S^1$ under multiplication.

Let $G$ be a connected and locally simply connected topological group with unit element $e$. Let $\pi : \tilde{G}_e \to G$ be a universal cover, and $\tilde{e}$ the universal element in the fiber $\pi^{-1}(e)$.

**(a) (Difficulty 20/10)**

Equip $\tilde{G}_e$ with a natural structure of a topological group with unit element $\tilde{e}$ for which $\pi$ becomes a homomorphism of topological groups.

**(b) (Difficulty 18/10)**

Show that a group structure with the above properties is unique.

**(c) (Difficulty 16/10)**

Construct an exact sequence of topological groups

$$1 \to \pi_1(G, e)^{\mathrm{op}} \to \tilde{G}_e \to G \to 1,$$

the topology on $\pi_1(G, e)$ being discrete.